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Stopped Unlimited Algebra Campaign

Certified Orientation Premises, a Coefficient-Field Failure, and Honest Nonresults from P485, P487, and P475

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Completed result. P486 exactly certifies, on the P473 root box, all orientation, branch-selection, pivot-localization, and nonzero-reference premises required by the proposed P485/P487 reductions.

Failed computations. P485 and P487 did not enter their Gröbner computations. Both failed while coercing an explicit $-4\sqrt{2}$ coefficient into $\mathbb{Q}$. This is an implementation-domain error, not a mathematical counterexample or no-go theorem.

Manual stop. The original fourteen-variable P475 lexicographic computation was manually interrupted after running for more than twelve hours without returning a basis. It yields no new scientific verdict.

Recoverable defect. For $\alpha = \sin(\pi/8)$, $\sqrt{2} = 2 - 4\alpha^2$. Hence the offending coefficient already lies in $\mathbb{Q}(\alpha)$. This repair is proved here but has not been executed through P485/P487.

Publication statement

This monograph records the local analytical and computational state of the P486–P485–P487–P475 campaign at manual interruption. It uses no network, Firecrawl, external AI, external audit, remote computation, laboratory record, or empirical physical evidence. A stopped or failed program is not silently promoted to a theorem.

Contents

Chapter 1

Executive summary

The campaign was launched sequentially to avoid competition between exact computer-algebra jobs on a 16-GB host. The execution order was

$$\text{P486} \longrightarrow \text{P485} \longrightarrow \text{P487} \longrightarrow \text{P475-unlimited}.$$

No application-level CPU, wall-time, or memory cap was imposed. The operating system and physical machine nevertheless remained real constraints.

Only P486 completed scientifically. It exports O188, the P473 Orientation-and-Localization Premise Certificate. The exact output proves:

- $\det C > 0$ uniformly on the P473 box;
- one declared 3×3 pivot for eliminating A, B, u is uniformly nonzero;
- the selected coefficient in the affine tangent parameter ρ is uniformly nonzero;
- $B > 0$, $L - b > 0$, and $\alpha > 0$;
- the exact standard-sector residual selects $125(L - b) - 36\alpha = 0$, not its negative companion;
- the induced metric rotation has determinant $+1$.

P486 does *not* prove the missing P485 assertion that the invariant axis belongs to the full causal equal-endpoint plane. Thus it does not yet turn the P484 complex phase face into an exact theorem.

P485 and P487 both failed before Gröbner-basis construction. The source requested `Poly(..., domain=QQ)` after the parity-basis transformation, but a coefficient remained as $-4\sqrt{2}$. The correct classification is [Implementation failure], not refutation. The obstruction is representational and repairable because $\sqrt{2} \in \mathbb{Q}(\alpha)$.

P475 then entered the original full fourteen-variable lexicographic Gröbner computation. It was manually interrupted at the user's request after more than twelve hours. No basis, elimination polynomial, or result JSON was returned. Its status is therefore [Stopped] and its mathematical question remains [Open].

Program	Execution status	Scientific status
P486	Completed	[Computer-assisted proof] premise certificate O188
P485	Failed before Gröbner	[Open]; no causal-axis verdict
P487	Failed before Gröbner	[Open]; no relation for L
P475 unlimited	Manually interrupted	[Open]; no basis and no no-go

The principal methodological result of this checkpoint is epistemic as well as algebraic: a long calculation has been stopped without turning incomplete machine work into evidence. The durable positive mathematics is P486 plus the new exact coefficient-closure identity O191.

Chapter 2

Inherited exact problem

2.1 The reduced three-slot operator problem

The campaign operates downstream of the declared finite three-slot process tester. For the compressed Hermitian process difference Δ, the inner value at a positive comb normalizer N is

$$F(N) = \frac{1}{2} \left\| N^{1/2} \Delta N^{1/2} \right\|_1 .$$

P473 certified, inside a rational thirteen-dimensional box, a positive solution of the structured Riccati equation

$$X_3 N X_3 = \frac{1}{4} \Delta N \Delta, \tag{2.1}$$

and used the exact causal block pattern and trace telescope to establish primal–dual attainment. The value lies in

$$0.5233281002710117 \leq L \leq 0.5233281002710717. \tag{2.2}$$

Root uniqueness is certified only inside that local real chart.

2.2 Parity form and the complex tangent question

The exact P484 parity representation splits the active Riccati system into three-dimensional blocks

$$X_+ N_0 X_+ = \frac{1}{4} C N_0 C^{\mathsf{T}}, \qquad X_- N_0 X_- = \frac{1}{4} C^{\mathsf{T}} N_0 C. \tag{2.3}$$

An antisymmetric imaginary causal direction is parametrized by one scalar ρ. Its linearized contact condition reduces to six affine equations

$$c_j(z)\rho + d_j(z) = 0, \qquad j = 0, \ldots, 5, \tag{2.4}$$

where z denotes the exact real Riccati variables and $\alpha = \sin(\pi/8)$.

Choosing a reference equation r with $c_r \neq 0$, consistency is equivalent to five polynomial identities

$$c_r d_j - c_j d_r = 0, \qquad j \neq r. \tag{2.5}$$

P485 was designed to test whether these five polynomials belong to the localized positive-branch Riccati ideal. This is the precise missing theorem; neither numerical near-zero values nor orientation alone suffice.

2.3 Exact coefficient field

All trigonometric input is generated by

$$\alpha = \sin(\pi/8), \qquad 8\alpha^4 - 8\alpha^2 + 1 = 0, \tag{2.6}$$

with

$$\sin(\pi/4) = 1 - 2\alpha^2, qquad \sin(3\pi/8) = 3\alpha - 4\alpha^3. \tag{2.7}$$

Thus the intended coefficient field is the degree-four algebraic field $\mathbb{Q}(\alpha)$, represented computationally by adjoining the polynomial (2.6) to rational polynomial equations.

Chapter 3

P486: completed orientation and localization certificate

3.1 Uniform determinant payments

P486 evaluated exact rational polynomial intervals over the complete P473 box. The key enclosures are

$$\begin{aligned} \det C &\in [0.04328307091427969,\ 0.04328307091427969], \\ \det P_{3,5,6} &\in [-0.15345958532886037,\ -0.1534595852158078], \\ c_r &\in [-0.03682747265490332,\ -0.03682747265474969], \\ B &\in [0.0687772109049226,\ 0.0687772109049826], \\ L - b &\in [0.1102128285210847,\ 0.1102128285212047]. \end{aligned} \tag{3.1}$$

The displayed decimals summarize stored rational endpoints; the sign tests were performed on the rational endpoints, not by floating rounding.

The nonzero pivot $P_{3,5,6}$ licenses exact Cramer/adjugate elimination of the three normalizer coordinates A, B, u on this local branch. The nonzero reference coefficient c_r licenses

$$\rho = -\frac{d_r}{c_r}. \tag{3.2}$$

These facts are prerequisites for P485 and P487; they do not themselves make the remaining equations vanish.

3.2 Positive standard branch

The standard antisymmetric sector gives the exact factorization

$$-\frac{B}{15625}(-125L + 125b + 36\alpha)(125L - 125b + 36\alpha) = 0. \tag{3.3}$$

On the certified box, $B > 0$, $L - b > 0$, and $\alpha > 0$. The factor $125(L - b) + 36\alpha$ is strictly positive, hence (3.3) selects

$$125(L - b) - 36\alpha = 0, \qquad b = L - \frac{36}{125}\alpha. \tag{3.4}$$

This is not a heuristic branch choice. It is an exact consequence of the factorization and certified signs.

3.3 Orientation theorem

Define

$$\mathcal{B} = X_-^{-1} \frac{C^\mathsf{T}}{2}.$$

The second identity in (2.3) gives

$$\mathcal{B} N_0 \mathcal{B}^\mathsf{T} = N_0. \tag{3.5}$$

Taking determinants yields $(\det \mathcal{B})^2 = 1$. P473 supplies $X_- > 0$, so $\det X_- > 0$; P486 supplies $\det C > 0$. Therefore

$$\det \mathcal{B} = +1. \tag{3.6}$$

This proves that $\mathcal{B}$ is an orientation-preserving isometry of the positive metric N_0. In odd dimension it has a fixed axis.

The exact boundary is essential: (3.6) proves existence of an invariant axis, but it does not prove that this axis satisfies the independent causal equal-endpoint constraint. P484 already supplied an exact counterexample to the claim that odd dimension plus metric orthogonality alone would force that membership. P485 remains necessary.

3.4 Verdict

[Computer-assisted proof] O188 is accepted in its declared scope. It pays every explicit orientation, branch, pivot, and reference-nonvanishing premise used by the proposed reduced computations. It does not prove the complex phase face, full-cone nonuniqueness, or any physical interpretation.

Chapter 4

P485: intended theorem and actual failure

4.1 Exact ideal-membership design

Let I_+ be generated by the nine post-pivot active Riccati residuals, the minimal polynomial (2.6), and the localizer equation

$$t \det P_{3,5,6} - 1 = 0. \tag{4.1}$$

Equation (3.4) has already been substituted. P485 asks whether each consistency polynomial in (2.5) reduces to zero modulo a Gröbner basis of I_+. A successful certificate, combined with $c_r \neq 0$, would produce one exact ρ. Together with strict positivity of P473 this would prove that a nontrivial local complex optimal segment exists in the declared finite comb and would refute uniqueness over the full complex causal cone.

That theorem was never reached.

4.2 Observed exception

During construction of the reduced polynomial system, the program called

```
Poly(expression, variables, domain=QQ).
```

At least one transformed expression contained the coefficient $-4\sqrt{2}$. SymPy therefore raised

```
CoercionFailed: expected ‘Rational‘ object, got -4*sqrt(2)
```

before the Gröbner-basis call. The stored status and traceback identify the failure point in `reduced_branch_system`.

The correct verdict is:

- no ideal basis was computed;
- no target remainder was computed;
- no causal-axis identity was proved;
- no counterexample was produced;
- no algebraic complexity no-go follows.

4.3 Why this is representational, not mathematical

The explicit square root enters through the orthonormal parity basis. It is not a genuinely new field generator. Indeed,

$$\alpha^2 = \sin^2(\pi/8) = \frac{1-\cos(\pi/4)}{2} = \frac{2-\sqrt{2}}{4}.$$

Consequently

$$\boxed{\sqrt{2} = 2 - 4\alpha^2}. \tag{4.2}$$

This defines O191, the coefficient-field closure map. After substituting (4.2), every such parity coefficient is again polynomial over $\mathbb{Q}$ in α, subject to (2.6).

Equation (4.2) is [Proven]. The repaired P485 computation is [Open] because the substitution and subsequent ideal computation have not been executed in this campaign.

Chapter 5

P487: structure-aware elimination and the same early failure

P487 was intended to improve the blind P475 workflow in three steps:

1. eliminate A, B, u through the certified nonzero pivot;
2. select the positive branch (3.4);
3. localize at the pivot and order L last in an exact lexicographic elimination.

This is mathematically preferable to attacking all fourteen variables at once. A returned polynomial solely in L would still require factor isolation at the P473 interval, irreducibility, and exact substitution before being called a minimal polynomial.

P487 used the same reduction helper as P485 and encountered the same $-4\sqrt{2}$ coercion failure before starting its lexicographic Gröbner calculation. Therefore it returned no basis and no relation in L.

The failure does not invalidate the structural reduction: O188 licenses the pivot and branch, and O191 gives an exact coefficient normalization. It does show that a program specification is incomplete until its coefficient domain is closed under every basis transformation actually used.

Chapter 6

P475 unlimited: manual interruption

P475 retained the original full polynomial system:

- fourteen variables;
- thirteen Riccati equations plus the algebraic equation (2.6);
- exact rational coefficients after direct sine substitution;
- lexicographic order with L last;
- no application-level resource limit.

It entered the stage `full_lex_groebner_running` with fourteen generators. The job remained there for more than twelve hours. At the user's request the encompassing execution session was sent an interrupt and exited with code 130. A process check after interruption found no remaining P475, P485, P487, or campaign runner.

No compressed basis file and no `FIN_Program_475_Unlimited_Results.json` were produced. Thus the manual stop means only:

this particular exact computation did not finish before interruption.

It does *not* mean that L is transcendental, that the ideal is positive-dimensional, that no elimination relation exists, or that exact elimination is impossible.

This result is consistent with the earlier 45-second, 3-GiB P475 test but is not logically stronger as a theorem: both are observations about workflows, not algebraic impossibility proofs.

Chapter 7

Epistemic ledger and nonclaims

Statement	**Status**	**Reason**
P473 local positive Riccati root and attainment	[Computer-assisted proof]	Inherited exact rational interval certificate
P486 signs, branch, pivot, and orientation	[Computer-assisted proof]	Exact symbolic identities and rational intervals
$\sqrt{2} = 2 - 4\alpha^2$	[Proven]	Elementary identity for $\alpha = \sin(\pi/8)$
P485 causal-axis ideal membership	[Open]	Program failed before basis construction
Exact complex phase face	[Open]	Causal equal-endpoint identity remains unpaid
P487 relation in L	[Open]	Program failed before elimination
P475 full lex result	[Open]	Manually interrupted; no basis returned
Minimal polynomial of L	[Open]	No isolated irreducible factor
Full complex optimizer uniqueness	[Open]	Strong evidence against; exact phase face absent
Selector / QW-2191	[Boundary]	No non-premise strict source
Dimensional physics	[Boundary]	No canonical time, length, action, mass, or energy unit
Laboratory realization	[Boundary]	No apparatus, calibration, custody, or event record
Legacy–strict role transfer	[Boundary]	Completion and transfer theorem absent

In particular, this campaign exports no new kernel theorem, no strict selector, no dimensional constant, no physical apparatus, no laboratory evidence, no legacy-to-strict completion, no physical-role transfer, no L_{total}, and no Standard Model, gravity, cosmological, or Theory-of-Everything closure.

Chapter 8

A safer restart specification

The present campaign is stopped and must not restart automatically. If a future explicit instruction reopens it, the smallest responsible sequence is:

8.1 P485R: coefficient closure before algebra

1. Apply the exact normalization $\sqrt{2} \mapsto 2 - 4\alpha^2$ after every parity-basis transformation.
2. Assert that every coefficient belongs to $\mathbb{Q}[\alpha, z]$ before constructing `Poly(..., domain=QQ)`.
3. Add a unit test that deliberately searches all expressions for `sqrt(2)` and algebraic-number atoms.
4. Serialize generator and target statistics before entering a basis computation.
5. Run a short bounded dry run and preserve its checkpoint even if the subsequent exact calculation is stopped.

8.2 P485M: modular scouting before characteristic zero

Reduce the rationalized ideal modulo several good primes, excluding primes that annihilate the certified pivot denominators. Measure dimension, leading monomials, degree growth, and target remainders. Agreement across primes is not a characteristic-zero proof, but disagreement can cheaply expose a faulty ordering or positive-dimensional branch before another long run.

8.3 P485C: check the five targets separately

Rather than computing more than needed, seek explicit ideal certificates

$$q_j = \sum_k h_{jk} g_k$$

for the five consistency polynomials q_j. A sparse syzygy or normal-form computation may be much cheaper than a complete elimination basis. Each target should produce its own durable result.

8.4 P487R: structure-aware value elimination

Only after the normalized reduced ideal passes exact construction tests should value elimination be attempted. Use graded order first, inspect dimension and Hilbert data, and convert to lexicographic form only if the branch is certified zero-dimensional. A returned factor must be isolated by the P473 interval and proved irreducible before promotion.

8.5 Checkpoint policy

Every future long calculation should emit a PDF-ready state packet at these milestones:

1. exact system constructed and coefficient audit passed;
2. branch and localizer verified against O188;
3. modular scouts completed;
4. each target remainder or certificate completed;
5. elimination factor isolated and independently replayed.

This policy addresses the practical risk that computational credits or wall time end before a final monograph exists.

Chapter 9

Conclusions

The stopped campaign contains one genuine advance and three honest nonresults.

The advance is P486: the local P473 solution is now surrounded by an exact orientation-and-localization premise certificate. The correct positive standard branch is fixed, the elimination pivot and tangent reference are nonzero, and the induced metric rotation has determinant $+1$.

The remaining causal-axis theorem was not decided. P485 and P487 were blocked by a coefficient representation error before algebraic elimination began. The error is exactly repairable inside the existing field by O191, $\sqrt{2} = 2 - 4\alpha^2$, but the repaired computations have not been run. P475 was manually interrupted and provides no mathematical verdict.

Therefore the deepest conclusion justified by this checkpoint is modest and precise: the orientation premises are certified, the computational path is repairable, and the central exact phase-face and value-elimination questions remain open. No background computation remains active, and no restart is authorized by this document.

Appendix A

Artifact map

Artifact	Role
`FIN_Program_486_Results.json`	Completed exact interval/symbolic certificate
`FIN_Program_485_Unlimited_Status.json`	Stored P485 exception and traceback
`FIN_Program_487_Unlimited_Status.json`	Stored P487 exception and traceback
`FIN_Program_475_Unlimited_Status.json`	Manual-stop marker and last active stage
`FIN_Unlimited_Research_Pipeline_State.json`	Corrected campaign terminal state
`FIN_Local_Research_Checkpoint_P486_P485_P487_P475_Stopped_State.json`	Machine-readable epistemic ledger
`fin_phase_exact_algebra.py`	Shared exact construction; contains the identified pre-repair domain defect
`fin_program_486.py`	Completed P486 implementation
`fin_program_485_unlimited.py`	P485 intended ideal-membership implementation
`fin_program_487_unlimited.py`	P487 intended structure-aware elimination
`fin_program_475_unlimited.py`	Original full lexicographic cross-check

Appendix B

Status vocabulary

[Proven] denotes an exact analytic consequence. [Computer-assisted proof] denotes a finite exact rational or outward-interval certificate whose artifacts are available locally. [Open] denotes a proposition not decided by the current artifacts. [Implementation failure] denotes a software or representation failure before the scientific decision point. [Stopped] denotes a manual interruption with no returned result. These categories are not interchangeable.